

Quantitative Methods

Multiple linear regression model

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + \dots + b_kX_{ki} + \varepsilon_i, i = 1, 2, \dots, n$$

Assumptions of multiple linear regression

1. **Linearity:** The relationship between the dependent variable and the independent variables is linear.
2. **Homoskedasticity:** The variance of the regression residuals is the same for all observations.
3. **Independence of errors:** The observations are independent of one another. This implies the regression residuals are uncorrelated across observations.
4. **Normality:** The regression residuals are normally distributed.
5. **Independence of independent variables:**
 - 5a. Independent variables are not random.
 - 5b. There is no exact linear relation between two or more of the independent variables or combinations of the independent variables.

Commonly used diagnostic plots

Scatterplots of dependent and independent variables are used to check if the assumptions of 'linearity' and 'independence of independent variables' have been violated.

Scatterplots of residuals are used to check if the assumptions of 'homoskedasticity' and 'independence of errors' have been violated.

A 'Q-Q' plot of residuals is used to check if the assumption of 'normality' has been violated.

Goodness of Fit

R-squared (coefficient of determination) measures the fraction of the total variation in the dependent variable that is explained by the independent variable.

$$R^2 = \frac{\text{explained variation}}{\text{total variation}}$$

Total variation = unexplained variation + explained variation

AIC and BIC: lower values of both measures are better.

AIC is preferred if the model is used for prediction purposes.

BIC is preferred when goodness of fit is the goal.

Joint F-test

$$F \text{ stat} = \frac{(\text{Sum of squares error restricted model} - \text{Sum of squares error unrestricted})/q}{\text{Sum of squares error unrestricted model}/(n - k - 1)}$$

where: q is the number of restrictions, i.e. the number of variables omitted

If the calculated F-stat exceeds the critical F-value, we can conclude that at least one of the omitted variables is statically significant.

General linear F-test

$$F \text{ stat} = \frac{\text{mean regression sum of squares}}{\text{mean squared error}} = \frac{MSR}{MSE}$$

If the calculated F-stat exceeds the critical F-value, we can conclude that at least one of the slope coefficients is different from 0.

Functional Form

Failures in Regression Functional Form	Explanation	Consequence
Omitted variables	One or more important variables are omitted from the regression	May lead to heteroskedasticity or serial correlation
Inappropriate form of variables	Ignoring a nonlinear relationship between the dependent and independent variable	May lead to heteroskedasticity
Inappropriate variable scaling	One or more regression variables may need to be transformed before estimating the regression.	May lead to heteroskedasticity or multicollinearity
Inappropriate data pooling	Regression model pools data from different samples that should not be pooled	May lead to heteroskedasticity or serial correlation

Violation of Regression Assumptions

Assumption	Violation	Issue	Detection	Correction
Homoskedastic error terms	Heteroskedastic error terms	Biased estimates of coefficients' standard errors	Visual inspection of residuals; Breusch-Pagan test	Revise model; use robust standard errors
	Serial correlation	Inconsistent estimates of coefficients and biased standard errors	Breusch-Godfrey test	Revise model; use serial-correlation consistent standard errors
Independence of observations				
Independence of independent variables	Multicollinearity	Inflated standard errors	Variance inflation factor	Revise model; increase sample size

High leverage point: a data point having an extreme value of an independent variable (X).

Outlier: a data point having an extreme value of the dependent variable (Y).

Detecting Influential Points

- **Leverage** (detects high leverage points): if leverage $> 3 \left(\frac{k+1}{n} \right)$: observation is potentially influential where k = number of independent variables
- **Studentized residual** (detects outliers):

If	Then
$ t_i > 3$	Flag observation as being an outlier
$ t_i > \text{critical value of the t-statistic with } n-k-2 \text{ degrees of freedom}$	Flag outlier observation as being potentially influential

Qualitative Independent Variables

- Dummy variables take on a value of 1 if a particular condition is true and 0 if that condition is false. To distinguish among n categories, the model must include n - 1 dummy variables.
- An intercept dummy adds to or reduces the original intercept if a specific condition is met. When the intercept dummy is 1, the regression line shifts up or down parallel to the base regression line.

$$Y_i = b_0 + d_0D_i + b_1X_i + \varepsilon_i$$

- A slope dummy allows for the slope of the regression line to change if a specific condition is met.

$$Y_i = b_0 + b_1X_i + d_1D_iX_i + \varepsilon_i$$

- It is also possible for a regression model to use both intercept and slope dummy variables.

$$Y_i = b_0 + d_0D_i + b_1X_i + d_1D_iX_i + \varepsilon_i$$

Qualitative Dependent Variables

- Qualitative dependent variables are outcome variables describing data that fit into categories (e.g. bankrupt or not bankrupt). A logistic transformation is applied when the model contains qualitative dependent variables. The logistic transformation is:

$$\ln[P/(1 - P)]$$

- The natural logarithm (ln) of the odds of an event happening is the log odds which is also called the logit function.
- In a logit model, slope coefficients are interpreted as the change in the log odds that the event happens per unit change in the independent variable, holding all other independent variables constant.

A likelihood ratio (LR) test is a method to assess the fit of logistic regression models. The test is similar to the joint F-test. It compares the fit of the restricted and unrestricted models.

Trend models

- Linear trend model: dependent variable changes at a constant rate with time. The independent variable is time: $Y = b_0 + b_1t + \varepsilon_t$, $t = 1, 2, \dots, T$.
- Log-linear trends work well in fitting time series that have exponential growth.
- An autoregressive model (AR) is a time series where a given variable is regressed on its own past values.

$$X_t = b_0 + b_1X_{t-1} + \varepsilon_t$$

- For AR models to work the time series must be covariance-stationary: Constant expected value, variance and covariance.
- Durbin-Watson does NOT work for AR models.
- Test whether the autocorrelations of the error term (error autocorrelations) differ significantly from 0. Test-stat = residual autocorrelation / standard error
- Compare the out-of-sample forecasting performance of forecasting models by comparing their root mean squared error (RMSE), which is the square root of the average squared error.
- Mean-reverting level is given by:

$$x_t = \frac{b_0}{(1 - b_1)}$$

- A random walk is a time series in which the value of the series in one period is the value of the series in the previous period plus an unpredictable random error.

$$x_t = x_{t-1} + E(\varepsilon)_t = 0$$

- With a random walk $b_1 = 1$, so MRL is undefined. This is called the unit root problem. Solution is to use first differencing.

Supervised machine learning algorithms: penalized regression, support vector machine (SVM), K-nearest neighbor (KNN), classification and regression trees (CART), random forest classifier, ensemble learning

Unsupervised machine learning algorithms: principal component analysis (PCA), K-means algorithm, hierarchical clustering

Neural networks: deep learning nets (DLNs), reinforcement learning (RL)

Data prep & wrangling involves data cleansing and data pre-processing. The steps vary based on whether we are working with structured data or unstructured data.

Data exploration includes three steps: exploratory data analysis (EDA), feature selection, and feature engineering.

Model training consists of three major tasks: method selection, performance evaluation, and model tuning.

The following metrics are used to evaluate a **confusion matrix**:

$$\text{Precision (P)} = TP / (TP + FP)$$

$$\text{Recall (R)} = TP / (TP + FN)$$

$$\text{Accuracy} = (TP + TN) / (TP + FP + TN + FN)$$

$$\text{F1 score} = (2 * P * R) / (P + R)$$

Economics

Bid-ask spread is impacted by:

- Currency pair involved
- Time of day
- Market volatility
- Size of transaction

Cross rates: $A/B = 2.0000/2.0006$ and $B/C = 4.0000/4.0008$

Implied A/C cross rate = $8.0000/8.0040$

Forward exchange rates are quoted in terms of points to be added to the spot exchange rate. Forward points represent the difference between the forward rate and the spot rate. If the points are positive (negative), the base currency is trading at a forward premium (discount).

International parity conditions

$$\text{Covered interest rate parity} = F_f / d = (S_f / d) \left(\frac{1 + i_f \left[\frac{\text{Actual}}{360} \right]}{1 + i_d \left[\frac{\text{Actual}}{360} \right]} \right)$$

Uncovered interest rate parity: Expected % change in spot rate $(P/B) \approx i_p - i_B$

- If covered interest rate parity and uncovered interest rate parity hold then forward rates are unbiased estimates of future spot rates.
- Ex ante purchasing power parity: Expected % change in spot rate $(P/B) \approx \Pi_P - \Pi_B$
- International Fisher effect assumes that uncovered interest rate parity and ex ante purchasing power parity hold. If so: $i_p - i_B = \Pi_P - \Pi_B$

Balance of payments and exchange rates

For the most part, countries that run persistent current account deficits will see their currencies weaken over time. Similarly, countries that run persistent current account surpluses will tend to see their currencies appreciate over time.

Monetary policy and Mundell-Fleming

- In the Mundell-Fleming model, monetary policy affects the exchange rate primarily through the interest rate sensitivity of capital flows, strengthening the currency when monetary policy is tightened and weakening it when monetary policy is eased. The more sensitive capital flows are to the change in interest rates, the greater the exchange rate's responsiveness to the change in monetary policy.
- Countries that pursue overly easy monetary policies will see their currencies depreciate over time.
- Under conditions of high capital mobility, countries that simultaneously pursue expansionary fiscal policies and relatively tight monetary policies should see their currencies strengthen over time.

Economic growth and investment decisions

$$P = GDP \left(\frac{E}{GDP} \right) \left(\frac{P}{E} \right)$$

P represents aggregate price (value) of stocks; E represents aggregate earnings.

$$\text{Growth accounting: } \frac{\Delta Y}{Y} = \frac{\Delta A}{A} = \alpha \frac{\Delta K}{K} + (1 - \alpha) \frac{\Delta L}{L}$$

Growth rate in potential GDP = Rate of technological change + Long-term growth rate of labor force + Long-term growth rate in labor productivity

Neo-classical model:

Growth rate of output per capita = $\frac{\theta}{1-\alpha}$

Growth rate of output = $\frac{\theta}{1-\alpha} + n$

- θ is growth rate of TFP
- α is the elasticity of output with respect to capital
- n is the growth rate of labor

Labor productivity:

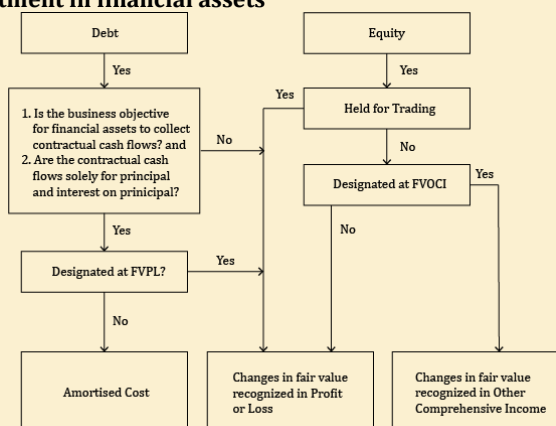
$$y = Y/L = A (K/L)^\alpha (L/L)^{1-\alpha} = Ak^\alpha$$

Financial Statement Analysis

Intercompany investments

	Financial assets	Associates	Business combinations	Joint ventures
Influence	Not significant	Significant	Controlling	Shared control
Typical % interest	Usually < 20%	Usually 20% to 50%	Usually > 50%	Varies
Financial Reporting	Classified as: • Fair value through profit or loss • Fair value through other comprehensive income • Amortized cost	Equity Method	Consolidation	IFRS: Equity method

Investment in financial assets



Investments in associates and joint ventures

An investment is considered an “associate company” when the investor has (or can exercise) significant influence, but not control, over the investee’s business activities. Equity method of accounting is required for investment in associates and joint venture.

- Investment is initially recorded at cost.
- Share of income (not dividends) recorded in investor’s I/S.
- Investment account reflected as single line item on B/S.
- Value of investment = beginning value + share of profit - share of dividends
- Investment classified as noncurrent asset on B/S.
- One-line consolidation.

Business combinations

Acquisition method

- Combine assets, liabilities, revenues and expenses of subsidiary with parent.
- Intercompany transactions are eliminated.
- For business combinations with less than 100% acquisition show non-controlling (minority) interests on balance sheet. U.S. GAAP says use full goodwill method. IFRS says either full goodwill or partial goodwill methods can be used.

Difference between full goodwill and partial goodwill

- Small company has net assets worth 90 but the overall company is valued at 100.
- Big company buys 80% of small company.
- Under full goodwill method a goodwill of 10 will be shown
- Under partial goodwill method, a goodwill of 8 will be shown.

Comparison of methods

	Equity method	Acquisition method (Partial goodwill)	Acquisition method (Full goodwill)
Assets	Net assets	All assets (partial goodwill)	All assets (full goodwill)
Liabilities		All liabilities	All liabilities
Equity	Net assets	More than net assets because of minority interest	Little more relative to partial goodwill method
Revenue / Exp.		Full consolidation	Full consolidation
N. I.	Same	Same	Same
Profit margin	More favorable	Less favorable	Less favorable
Leverage	Most favorable because liabilities are low	Less favorable than equity method	Slightly better than partial goodwill method
ROE and ROA	Most favorable	Worse than equity method but slightly better than full goodwill method	Least favorable

Restricted stock

Common stock granted to employees but is subject to selling and other restrictions. Its fair value is the market price of the underlying share at the grant date, which is expensed over the vesting period if vesting conditions are met or likely to be met. Settlement occurs simultaneously with vesting, as the RSUs convert to common shares.

Financial reporting vs. tax treatment of share-based compensation

	Financial reporting: stock-based compensation expense	Tax return: deduction for stock-based compensation
Timing	Over the vesting period	At settlement
Amount	Grant-date fair value	Share price on the settlement date (RSUs) Intrinsic value at exercise (options)

IFRS and US GAAP treatment of tax windfalls and shortfalls

	IFRS	US GAAP
Share price on settlement date > share price on grant date (excess tax benefit or tax windfall)	Gain recognized directly in stockholders’ equity.	Decrease in income tax expense on the income statement.
Share price on settlement date < share price on grant date (tax shortfall)	Loss recognized directly in stockholders’ equity.	Increase in income tax expense on the income statement.

Forecasted basic shares outstanding, end of period = Basic shares outstanding, beginning of period + RSUs vested and/or share options exercised + share issuances from secondaries, acquisitions, etc. - share repurchases

Valuation considerations: Share-based compensation should be considered when doing company valuation. The valuation model needs to be modified to account for the effect of:

- dilution from outstanding but unvested share-based awards and
- dilution from future share-based awards

Defined contribution pension plans

Expense every period. Unpaid expense at end of period is shown as a liability.

Defined benefit pension plans

- Both IFRS and U.S. GAAP require funded status to be reported on the balance sheet.
- Funded status** = Fair Value of plan assets – Pension obligation.
- Reported as net pension liability or net pension asset.

Periodic pension costs recognized in P&L vs. OCI

IFRS component	IFRS recognition	U.S. GAAP component	U.S. GAAP recognition
Service costs	Recognized in P&L.	Current service costs	Recognized in P&L.
		Past service costs	Recognized in OCI and subsequently amortized to P&L over the service life of employees.
Net interest income / expense	Recognized in P&L as the following amount: Net pension liability or asset $\times$ interest rate	Interest expense on pension obligation	Recognized in P&L.
		Expected return on plan assets	Recognized in P&L as the following amount: Plan assets $\times$ expected return.
Re-measurement: Net return on plan assets and actuarial gains and losses	Recognized in OCI and <u>not</u> subsequently amortized to P&L.	Actuarial gains and losses including differences between the actual and expected returns on plan assets	Recognized immediately in P&L or, more commonly, recognized in OCI and subsequently amortized to P&L using the corridor or faster recognition method.
	Net return on plan assets = Actual return – (Plan assets $\times$ Interest rate). Actuarial gains and losses = Changes in a company's pension obligation arising from changes in actuarial assumptions.		Difference between expected and actual return on assets = Actual return – (Plan assets $\times$ Expected return). Actuarial gains and losses = Changes in a Company's pension obligation arising from changes in actuarial assumptions.

- Current service cost: amount by which pension obligation increases as a result of employees' service in the current period.
- Past service cost: amount by which pension obligation relating to employees' service in prior periods changes as a result of plan amendments or a plan curtailment.

Valuation considerations for post-employment benefits

- Underfunded plan:** treated like debt because the company is obligated to make benefit payments.
- Overfunded plan:** ignored in valuation because plan assets are solely for paying benefits; they cannot be withdrawn and distributed to shareholders.
- Future service costs:** deducted from free cash flow in a discounted cash flow model.

Multinational operations: Translation methods

Determine an entity's functional currency:

- The currency that influences sales prices.
- The currency of the country whose competitive forces and regulations mainly determine the sales prices.
- The currency that mainly influences labor, material, and other costs.
- The currency in which funds for financing are generated.
- The currency in which receipts from operating activities are retained.

Current rate method: Foreign subsidiary's functional currency is DIFFERENT from the parent's presentation currency. Foreign entity's foreign currency financial statements are translated into the parent's presentation currency using the following procedures:

- All assets and liabilities are translated at the current exchange rate at the balance sheet date.
- Stockholders' equity accounts are translated at historical exchange rates.
- Revenues and expenses are translated at the exchange rate that existed when the transactions took place.
- Translation adjustment is reflected in equity.

Temporal method: Foreign subsidiary's functional currency is the SAME as the parent's presentation currency. Monetary assets and liabilities are translated at the current exchange rate. Non-monetary assets and liabilities measured at historical cost are translated at historical exchange rates. Non-monetary assets and liabilities measured at current value are translated at the exchange rate at the date when the current value was determined.

- Stockholders' equity accounts are translated at historical exchange rates.
- Revenue and Expenses. Revenues and expenses, other than those expenses related to non-monetary assets, are translated at the exchange rate that existed when the transactions took place. Expenses related to non-monetary assets, such as cost of goods sold (inventory), depreciation (fixed assets), and amortization (intangible assets), are translated at the exchange rates used to translate the related assets.
- Translation gain/loss on income statement.

If the value of assets to be translated is greater than the value of liabilities to be translated then we have a net asset exposure. With the current rate there will almost always be a net asset exposure.

Translation in a hyperinflationary economy

U.S. GAAP requires use of the temporal method.

IFRS require financial statements to be first restated for inflation and then inflation adjusted financial statements be translated at the current exchange rate.

Purchasing power gains/losses that arises from holding monetary assets and monetary liabilities are included in net income (net monetary assets $\rightarrow$ purchasing power loss; net monetary liabilities $\rightarrow$ purchasing power gain)

Basel III framework

- Minimum capital requirement:** A bank must have sufficient equity capital to absorb the loss in value of assets in a financial crisis.
- Minimum liquidity:** A bank must hold enough high-quality liquid assets to cover its liquidity needs in a 30-day liquidity stress scenario.
- Stable funding:** A bank should have adequate amount of stable funding relative to the bank's liquidity needs over a one-year horizon.

CAMELS approach

- Capital adequacy** – Does the bank have sufficient capital given its assets?
- Asset quality** – What is the quality of the bank's financial assets?
- Management capabilities** – Is the management effective? What is its track record?
- Earnings** – What is the level of earnings? What is the quality of earnings? Are earnings trending up or down?
- Liquidity**: How strong is the liquidity position of the bank? What are the sources of funding? Are the sources of funding stable?
- Sensitivity to market risk**: How sensitive are the bank's earnings to market risk?

Key ratios used to analyze an insurance company

Ratio	Formula	What is Measured
Loss and loss adjustment expense ratio	$\frac{\text{Loss expense} + \text{loss adjustment expense}}{\text{Net premiums earned}}$	Success in estimating risks insured
Underwriting expense ratio	$\frac{\text{Underwriting expense}}{\text{Net premiums written}}$	Efficiency of money spent in obtaining new premiums
Combined ratio	Loss and loss adjustment expense ratio + Underwriting expense ratio	Overall efficiency of an underwriting operation
Dividends to policyholders (shareholders) ratio	$\frac{\text{Dividends to policyholders (shareholders)}}{\text{Net premiums earned}}$	Liquidity
Combined ratio after dividends	Combined ratio + Dividends to policyholders (shareholders) ratio	Efficiency of money spent in obtaining new premiums

Quality of financial reports

- The lowest-quality reports contain information that is pure fabrication.
- High-quality reports contain information that is relevant, complete, neutral, and free from error.

High quality earnings are sustainable and represent returns equal to or in excess of the company's cost of capital.

Quantitative tools to assess the likelihood of misreporting

Beneish model: Uses eight variables to detect earnings manipulation.

M-score = Score indicating probability of earnings manipulation; normally distributed random variable with a mean of 0 and a standard deviation of 1.

Decomposition of ROE (DuPont analysis)

ROE = return on assets × leverage = net profit margin × asset turnover × leverage

ROE = EBIT margin × tax burden × interest burden × asset turnover × leverage

Accrual ratios

Balance sheet accruals ratio for time $t = (\text{NOA}_t - \text{NOA}_{t-1}) / [(\text{NOA}_t + \text{NOA}_{t-1})/2]$

Cash flow accruals ratio for time $t = [\text{NI}_t - (\text{CFO}_t + \text{CFI}_t)] / [(\text{NOA}_t + \text{NOA}_{t-1})/2]$

Corporate Issuers

Divided policy theories

MM's dividend irrelevance theory: Assuming perfect capital markets (no taxes, no transaction costs and symmetric information), dividend policy does not matter.

The Bird in the Hand Argument: Investors prefer a dollar of dividends to a dollar of potential capital gains from reinvesting earnings because they view dividends as less risky. Under this assumption, a company that pays dividends will have a lower cost of equity capital than an otherwise similar company that does not pay dividends; the lower cost of equity should result in a higher share price.

Tax Argument: In a country that taxes dividends at higher rates than capital gains, taxable investors should prefer companies that pay low dividends and reinvest earnings in profitable growth opportunities.

Signaling content of dividends: Managers/board (insiders) know more than outside investors (asymmetric information). Board and management may use dividends to signal to investors how the company is really performing.

- Dividend initiation or increase is generally seen as a positive signal and stock prices will probably rise in the short-run.
- If revised dividends are not sustainable, the stock price will fall.
- Decrease/omission is generally seen as a negative signal.

Effective tax rates on dividends

Double taxation: Effective tax rate = corporate tax rate + $(1 - \text{corporate tax rate})(\text{individual marginal tax rate on dividends})$

Tax imputation system: Taxes on dividends are effectively levied only at the shareholder level.

Split rate system: Different tax rate for retained earnings vs. earnings that will be distributed.

Payout policies

Stable dividend: dividends are based on long term forecast of sales. Dividends are increased when earnings have increased to a sustainably higher level.

Constant dividend payout ratio: the company decides on a dividend payout ratio and then applies this ratio to current earnings to calculate dividend amount.

Dividend vs. share repurchase decision

Factors to consider when evaluating share repurchases:

- Potential tax advantages.
- Share price support/signaling that the company considers its shares a good investment.
- Added managerial flexibility.
- Offsetting dilution from employee stock options.
- Increasing financial leverage.

Analysis of dividend safety

- Traditional measure: dividend payout ratio and its inverse, the dividend coverage ratio.
- A more comprehensive measure: FCFE coverage ratio = $\text{FCFE} / (\text{dividends} + \text{share repurchases})$

Corporate governance

Corporate ownership structures can be classified as: dispersed, concentrated, or a hybrid of the two. Different kinds of conflicts exist depending on the ownership structure and voting power.

Two typical structures for a company's board of directors are: one-tier board, and two-tier board. CEO duality exists when the chief executive officer (CEO) also serves as the chairperson of the board.

To evaluate a company's board, we look at the following: board of director's structure, board independence, board skills and experience, and board composition.

ESG integration

While evaluating ESG factors, analysts first need to evaluate if information is material. They should also consider their investment horizon.

Equity vs. fixed-income security analysis

- Equity analysis: ESG integration is used to both identify potential opportunities and mitigate downside risk. Analysts adjust financial model variables such as cost of capital to reflect ESG factors.

- Fixed-income analysis: ESG integration is generally focused on mitigating downside risk. Analysts usually do not focus on potential opportunities. The credit assessment may vary depending on maturity.

Estimating Cost of Debt

For a company with publicly traded debt and no embedded options → YTM on the company's existing debt with longest maturity can be taken as cost of issuing new debt

For non-traded debt → synthetic credit rating and matrix pricing can be used to estimate cost of debt

Estimating the Equity Risk Premium

Historical equity risk premium estimate: the mean value of the differences between broad-based equity-market-index returns and government debt returns over a given time period

Forward-looking estimates:

- Survey estimates: Ask experts for their opinion.
- Dividend discount models: $ERP = \text{market dividend-yield} + \text{consensus long-term earnings growth rate} - \text{current long-term government bond yield}$
- Macroeconomic modeling: $ERP = [DY + \Delta(P/E) + i + g - \Delta S] - E(r_f)$

Estimating the Required Return on Equity

Dividend Discount Models: $r_e = \frac{D_1}{P_0} + g$

Bond Yield Plus Risk Premium Approach: cost of equity = YTM on the company's long-term debt + risk premium

Risk-Based Models:

- CAPM: $r_e = r_f + \beta (ERP)$
- Fama-French:
 - Three-factor: $r_e = r_f + \beta_1 ERP + \beta_2 SMB + \beta_3 HML$
 - Five-factor: $r_e = r_f + \beta_1 ERP + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA$

Estimating Cost of Equity for Private Companies

Expanded CAPM: $r_e = r_f + \beta_{\text{peer}} (ERP) + \text{size premium} + \text{industry risk premium} + \text{specific-company risk premium}$

Build-Up Approach: $r_e = r_f + ERP + \text{size premium} + \text{specific-company risk premium}$

International considerations

- $ERP = ERP \text{ for a developed market} + (\lambda \times \text{Country risk premium})$ where:
 - λ is the level of exposure of the company to the local country
 - Country risk premium = Sovereign yield spread $\times \sigma_{\text{Equity}} / \sigma_{\text{Bond}}$
- International CAPM: $E(r_e) = r_f + \beta_G (E(r_{gm}) - r_f) + \beta_C (E(r_C) - r_f)$ where:
 - r_{gm} = risk premium of a global index
 - r_C = risk premium of wealth-weighted foreign currency index

Motivations for corporate structural change:

	Investment Actions	Divestment Actions	Restructuring Actions
Issuer-specific motivations	- Realize synergies - Increase growth - Improve capabilities or secure resources - Opportunity to acquire an undervalued target	- Focus operations and business lines - Valuation - Liquidity needs - Regulatory requirements	- Improve returns on capital - Financial challenges, including bankruptcy and liquidation
Top-down drivers	- High security prices - Industry shocks		

Preliminary valuation of the target

Comparable company analysis: Calculate various relative value measures based on the current market price of comparable companies. Eg: EV/EBITDA, P/S, P/E, P/CF, etc. Review descriptive statistics (mean, median, and range) for the metrics chosen and apply judgment and experience to estimate company value. This method is often used to assess the value of targets in spin offs rather than acquisitions. This is because in acquisitions, acquirers pay a premium for control and the acquisition price typically exceeds the price estimated by this method.

Comparable transaction analysis: Calculate various relative measures like P/E, P/CF, P/B based on prices actually paid in recent and relevant takeover transactions (not current market prices). Review descriptive statistics (mean, median, and range) for the metrics chosen and apply judgment and experience to estimate company value.

Unlike comparable company analysis, the valuation multiples in comparable transaction analysis include takeover premiums, because they reflect historical acquisitions.

Premium paid analysis: Takeover premium = $\frac{(DP-SP)}{SP}$

where: DP = deal price per share of the target company and SP = stock price of the target company

Pro forma income statement

Pro Forma Income Statement (Acquisition) Modeling	
Revenue	1. Combine acquirer and target revenues. 2. Add revenue synergies or subtract the cost of incompatible activities (dis-synergies).
Operating expenses	1. Combine acquirer and target operating expenses. 2. Subtract cost synergies or add the cost of incompatible activities (dis-synergies).
Depreciation and amortization	1. Combine acquirer and target depreciation and amortization. 2. Add amortization of acquired intangible assets.
Other expense or income	1. Combine acquirer and target other expense or income.
Interest expense	1. Start with current acquirer interest expense. 2. Add increased interest from new debt issuance and revised interest rate.
Income taxes	1. EBT-weighted average of tax rates of acquirer and target; estimate usually provided by issuer
Shares outstanding	1. Start with current acquirer shares outstanding 2. Add shares from any share issuance

WACC: restructurings change can affect the issuer's WACC as costs of capital and weights are impacted.

Equity

Discounted dividend valuation

Dividend discount model

$$V_0 = \left[\sum_{t=1}^n \frac{D_t}{(1+r)^t} \right] + \frac{P_n}{(1+r)^n}$$

$$V_0 = \frac{D_0 \times (1+g)}{r-g}$$

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{1-b}{r-g}$$

$$\frac{P_0}{E_0} = \frac{D_0(1+g)/E_0}{r-g} = \frac{(1-b)(1+g)}{r-g}$$

$$V_0 = \frac{E_1}{r} + PVGO$$

$$\frac{P_0}{E_1} = \frac{1}{r} + \frac{PVGO}{E_1}$$

The H-model

$$V_0 = \frac{D_0(1+g_L)}{r-g_L} + \frac{D_0H(g_S-g_L)}{r-g_L}$$

Three-stage dividend discount models

- First version: company is assumed to have 3 distinct stages of growth and the growth rate of second stage is typically constant.
- Second version: the middle stage is similar to the first stage in the H-model. Dividends grow at a constant high rate in the first stage, decline linearly in the second stage, and grow at sustainable constant growth rate in the third stage.

Free cash flow valuation

- $FCFF = NI + Dep + Int(1 - \text{tax rate}) - FCInv - WCInv$
 $WCInv = \text{Change in working capital, excluding cash and short term debt}$
 $FCInv = \text{Change in gross fixed assets}$
- $FCFF = CFO + Int(1 - \text{tax rate}) - FCInv$
 If interest is not categorized in CFO, then do not add it back.
- $FCFF = EBIT(1 - \text{tax rate}) + dep - FCInv - WCInv$
- $FCFF = (EBITDA - dep)(1 - \text{tax rate}) + dep - FCInv - WCInv$
 $FCFF = EBITDA(1 - \text{tax rate}) + dep(\text{tax rate}) - FCInv - WCInv$
- $FCFE = FCFF - Int(1 - \text{tax rate}) + \text{net borrowing}$
 $FCFE = NI + NCC - FCInv - WCInv + \text{net borrowing}$

Cases in which FCFF-based valuation is preferred to FCFE-based valuation:

- A levered company with negative FCFE
- A levered company with changing capital structure

On historical basis, FCFF growth may be more representative of fundamentals than FCFE growth. On forward looking basis, given changing capital structure, the required return on equity may be more sensitive to changes in capital structure than the changes in WACC.

$$\text{Value of operating assets} = \sum_{t=1}^{\infty} \frac{FCFF_t}{(1+WACC)^t}$$

$$\text{Value of equity} = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1+r)^t}$$

Value of firm = Value of operating assets + Value of non-operating assets

Equity value = Firm value - Market value of debt

Enterprise value / EBITDA

Method based on forecasted fundamentals: Compute ratio based on company fundamentals and compare with ratio based on actual market price. If the ratio based on actual market price is low, then the stock is undervalued.

Method of comparables

Determine benchmark value:

- median industry P/E
- average industry P/E
- average of own past P/Es

If firm multiple > benchmark implies overvaluation and vice versa.

Advantages and drawbacks of using P/E for valuation

- Earnings are the key driver of value.
- P/E ratio is widely recognized and used by investors.
- Differences in P/Es may be related to differences in long-run average returns.
- Negative EPS → P/E would be meaningless.
- Earnings often contain volatile, transient components.
- Management's choice of accounting policies may taint EPS's value.

Rationales for the use of P/B

- Can be used when EPS is negative.
- BVPS is more stable than EPS (especially when EPS is volatile).
- More useful for valuing companies with liquid assets (banks, etc).
- Can be used when company is not a going concern.
- Differences in P/B may relate to differences in long run returns.

Drawbacks of using P/B

- Assets creating value (e.g., human capital) not on balance sheet.
- Can be misleading when companies have different level of assets.
- Accounting effects (e.g., R&D) can compromise BV.
- BV is generally not reflective of market value.

Rationales for the use of P/S

- Sales are less subject to accounting manipulation than EPS or BVPS.
- Sales are positive even when EPS is negative.
- Sales are more stable than EPS.
- More useful in valuing mature, cyclical and zero income companies.
- Differences in P/S may relate to differences in long-run returns.

Drawbacks of using P/S

- Might have high sales growth but low earnings and CFO.
- P/S does not capture differences in cost structures.
- Does not capture different capital structures.
- Room for manipulating revenues remains.

Rationales for the use of P/CF

- Less subject to manipulation.
- More stable than EPS.
- Addresses the issue of accounting differences.
- Differences in P/CF may relate to differences in long-run returns.

Drawbacks of using P/CF

- Earnings plus noncash charges, definition of cash flow ignores the changes in working capital; could use FCFE, but:
- FCFE is more volatile and more frequently negative.

Residual income valuation

- Residual income = Net income - (Equity capital x Cost of equity)
- Residual income = EBIT (1 - Tax rate) - (Total capital x WACC)
- Economic value added (EVA) = NOPAT - (C% x TC)
 NOPAT and TC are adjusted
- MVA = Market Value - Book Value
- $V_0 = B_0 + \sum_{t=1}^{\infty} \frac{RI_t}{(1+r)^t} = B_0 + \sum_{t=1}^{\infty} \frac{E_t - rB_{t-1}}{(1+r)^t}$
- $V_0 = B_0 + \sum_{t=1}^{\infty} \frac{(ROE_t - r)B_{t-1}}{(1+r)^t}$

If residual income continues indefinitely at a positive level:

$$V_0 = B_0 + \frac{ROE - r}{r - g} B_0$$

If at the end of time horizon T, a certain premium over book value ($P_T - B_T$) exists for the company:

$$V_0 = B_0 + \sum_{t=1}^T \frac{(E_t - rB_{t-1})}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

$$V_0 = B_0 + \sum_{t=1}^T \frac{(ROE_t - r)B_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

If residual income fades over time:

$$V_0 = B_0 + \sum_{t=1}^{T-1} \frac{(E_t - rB_{t-1})}{(1+r)^t} + \frac{E_T - rB_{T-1}}{(1+r-\omega)(1+r)^{T-1}}$$

Persistence factor, ω , is between 0 and 1

1 → residual income will not fade

0 → residual income will not continue after the initial forecast horizon.

Differences between private and public companies: can be categorized into company-specific and stock-specific factors.

- Company-specific factors:** stage in lifecycle, size, concentration of ownership, limited disclosures and overlap of shareholders & management.
- Stock-specific factors:** liquidity, concentration of control and potential agreements restricting liquidity.

Private Equity Valuation

Reasons for private company valuations: Transactions, Compliance and Litigation.

Approaches to private company valuation: Income Approach, Market Approach and Asset-Based Approach.

Methods used in the income approach: Free Cash Flow, Capitalized Cash Flow and Excess Earnings Method.

- Free cash flow method: Value of the firm = PV of expected future cash flows + PV of terminal value.
- Capitalized cash flow method: Value of the firm $V_f = \frac{FCFF_1}{WACC - g_f}$ or $V_e = \frac{FCFE_1}{r - g}$
- Excess earnings method: Value of the firm = Value of intangible assets + Value of fixed assets + Value of working capital where
Value of intangible assets = $\frac{RI(1+g)}{r - g}$

Methods used in a market approach: Guideline Public Company Method (GPCM), Guideline Transaction Method (GTM), and Prior Transaction Method (PTM).

Valuation discounts and premiums

Discount for lack of control = DLOC = $1 - [1 / (1 + \text{control premium})]$

A discount for lack of marketability (DLOM) is an amount or percentage deducted from the value of an ownership interest to reflect the relative absence of marketability.

Fixed Income

Spot rate and forward rate

Forward rate: interest rate that is determined today for a loan that will be initiated in a future time period.

$f_{A,B-A}$ forward rate of a loan initiated A years from today with maturity of B – A years.

$F_{A,B-A}$: forward contract price at time A for a zero-coupon bond with maturity B – A years and unit principal.

Forward pricing model: $DF_B = DF_A \times F_{A,B-A}$

Relationship between spot rate and forward rates: The spot rate for a given maturity can be expressed as a geometric average of the short-term rate and a series of forward rates.

Forward rate model:

$$[1 + z_B]^B = [1 + z_A]^A [1 + f_{A,B-A}]^{B-A}$$

When the spot curve is upward (downward) sloping → the forward curve will lie above (below) the spot curve.

Riding the yield curve or rolling down the yield curve

If a trader does not believe that the yield curve will change its level and shape over an investment horizon, he will buy bonds with a maturity longer than the investment horizon. This strategy is called riding the yield curve or rolling down the yield curve.

Spreads

- The Swap spread is the spread paid by the fixed-rate payer of an interest rate swap over the rate of the “on-the-run” (most recently issued) government security with the same maturity as the swap.
- The Z-spread is the constant basis point spread that would need to be added to the implied spot yield curve so that the discounted cash flows of a bond are equal to its current market price.
- The TED spread is calculated as the difference between MRR and the yield on a T-bill of matching maturity. The TED spread is an indicator of perceived credit risk in the general economy.
- The MRR-OIS spread is an indicator of the risk and liquidity of money market securities. It is the difference between MRR and the overnight indexed swap (OIS) rate. An OIS is an interest rate swap in which the periodic floating rate of the swap is equal to the geometric average of an overnight rate (or overnight index rate) over every day of the payment period. The index rate is

typically the rate for overnight unsecured lending between banks—for example, the federal funds rate for US dollars.

Traditional term structure theories

Unbiased expectations

- Also called pure expectations theory.
- Forward rate is an unbiased predictor of the future spot rate.
- Implication: buying a bond with a maturity of five years and holding it for three years has the same expected return as buying a three-year bond or buying a series of three one-year bonds.
- Can be used to explain any shape of the yield curve.
- Does not consider risk.

Local expectations

- More rigorous than the unbiased expectations theory.
- Expected return for every bond over short time periods is the risk-free rate.
- Assumption: bond pricing does not allow for traders to earn arbitrage profits.
- No risk premium for short time periods.
- Theory accommodates risk premium for long time periods.

Liquidity preference theory

- Liquidity preference theory asserts that liquidity premiums exist to compensate investors for the added interest rate risk they face when lending long term and that these premiums increase with maturity.
- Given an expectation of unchanging short-term spot rates, liquidity preference theory predicts an upward-sloping yield curve.
- Forward rate provides an estimate of the expected spot rate that is biased upward by the amount of the liquidity premium.
- Existence of liquidity premiums implies that the yield curve will typically be upward sloping.

Segmented markets theory

- Yields are solely a function of the supply and demand for funds of a particular maturity.
- Each maturity sector is considered a segmented market.
- The theory is consistent with a world where there are asset/liability management constraints, either regulatory or self-imposed.

Preferred habit theory

- The preferred habitat theory is similar to the segmented markets theory in proposing that many borrowers and lenders have strong preferences for particular maturities, but it does not assert that yields at different maturities are determined independently of each other.
- If the expected additional returns to be gained become large enough, institutions will be willing to deviate from their preferred maturities or habitats.
- Agents and institutions will accept additional risk in return for additional expected returns.

Bond's exposure to factors driving the yield curve

A bond's exposure to factors driving the yield curve can be measured and managed using:

- Effective duration: Measures the sensitivity of a bond's price to a small parallel shift in the benchmark yield curve.
- Key-rate duration: Measures a bond's sensitivity to a small change in a benchmark yield curve at a specific maturity segment.
- Sensitivity to parallel, steepness and curvature movements: Yield curve movements are decomposed into three distinct categories. Sensitivity to each category is calculated separately.

Effect of macroeconomic factors on bond returns and bond risk premium

Bond risk premium (also called term premium or duration premium) is the expected excess return of a default-free long-term

bond less the expected excess return of an equivalent short-term bond, or the one-period risk-free rate.

Many macroeconomic factors such as inflation, economic growth and monetary policy influence bond pricing and required returns.

Monetary policy: During economic expansions, monetary authorities raise benchmark rates to control inflation which results in a bearish flattening. During economic recessions, monetary authorities cut benchmark rates to stimulate economic activity which results in a bullish steepening. Central banks can also use asset purchases to influence the bond risk premium.

Other factors that influence the bond risk premium are:

- Fiscal policy
- Maturity structure of debt
- Investor demand

Pricing a bond using a binomial tree

Backward Induction: Start at maturity, fill in those values, and work back from right to left to find the bond's value at the desired node.

$$\text{Bond value at a node} = \frac{C + (0.5 \times V_H + 0.5 \times V_L)}{1 + i}$$

Term structure models

CIR model: $dr = a(b - r)dt + \sigma \sqrt{r} dz$

- assumes that interest rates are mean reverting and volatility increases as rates increase.

Vasicek model: $dr = a(b - r)dt + \sigma dz$

- interest rates are calculated assuming that volatility remains constant over the period of analysis. As with the CIR model, there is only one stochastic driver of the interest rate process. Both the Vasicek and CIR models assume a single factor, the short-term interest rate, r . The main disadvantage of the Vasicek model is that it is theoretically possible for the interest rate to become negative.

Ho-Lee model: $drt = \theta dt + \sigma dz$

- It is an arbitrage-free model and can be calibrated to closely match the observed term structure. Arbitrage-free models are frequently used to value bonds with embedded options.
- A benefit of arbitrage-free term structure models is that they are calibrated to the current term structure. In contrast, equilibrium term structure models frequently generate term structures that are inconsistent with current market data.
- Shaping risk is defined as the sensitivity of a bond's price to the changing shape of the yield curve.

The Kalotay-Williams-Fabozzi (KWF) model: $d \ln(rt) = \theta dt + \sigma dz$

- The KWF model is similar to the Ho-Lee model - it assumes constant drift, no mean reversion, and constant volatility. However, we use the log of the short rate as the dependent variable.
- The advantage of modeling the log of the short rate is that it prevents interest rates from becoming negative.

Model	Type	Short Rate	Drift Term	Volatility
CIR	Equilibrium	dr_t	Mean reversion at speed k	Varies with $\sqrt{r_t}$
Vasicek	Equilibrium	dr_t	Mean reversion at speed k	Constant
Ho-Lee	Arbitrage free	dr_t	Time dependent	Constant
KWF	Arbitrage free	$d \ln(rt)$	Time dependent	Constant

Valuation of bonds with embedded options

- Value of callable bond = Value of straight bond – Value of issuer call option
- Value of issuer call option = Value of straight bond – Value of callable bond
- Value of puttable bond = Value of straight bond + Value of investor put option
- Value of investor put option = Value of puttable bond – Value of straight bond
- The value of any embedded option increases with interest rate volatility.
- The value of an embedded call option is higher if the yield curve is downward sloping.
- The value of an embedded put option is higher if the yield curve is upward sloping.

$$\text{Effective duration} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta \text{Curve}) \times (PV_0)}$$

- Cash $\rightarrow$ effective duration: 0
- Zero-coupon bond $\rightarrow$ effective duration $\approx$ Maturity
- Fixed-rate bond $\rightarrow$ effective duration $<$ Maturity
- Callable bond $\rightarrow$ effective duration $\leq$ Duration of straight bond
- Puttable bond $\rightarrow$ effective duration $\leq$ Duration of straight bond
- Floater (MRR flat) $\rightarrow$ effective duration $\approx$ Time (in years) to next reset
- The effective duration of a callable bond cannot exceed that of the straight bond.
- One-sided up-duration (One-sided down-duration) measures the sensitivity of a bond's value to an increase (decrease) in interest rates.

$$\text{Effective convexity} = \frac{(PV_-) + (PV_+) - [2 \times (PV_0)]}{(\Delta \text{Curve})^2 \times (PV_0)}$$

Callable bonds: Negative convexity at low interest rates; Similar to straight bonds at high rates.

Puttable bonds: Similar to straight bonds at low rates; Positive but low convexity at high rates

Value of capped floater = Value of straight bond – Value of embedded cap

Value of floored floater = Value of straight bond + Value of embedded floor

Conversion value = Underlying share price $\times$ Conversion ratio

Market conversion premium per share = Market conversion price – Underlying share price

Market conversion price = convertible bond price / conversion ratio

Market conversion premium ratio = market conversion premium per share / share price

Premium over straight value = (convertible bond price / straight value) – 1

Value of convertible bond = Value of straight bond + Value of call option on the issuer's stock

Value of callable convertible bond = Value of straight bond + Value of call option on the issuer's stock – Value of issuer call option

Value of callable puttable, convertible bond = Value of straight bond + Value of call option on the issuer's stock – Value of issuer call option + Value of investor put option

Expected exposure is the maximum amount of money that an investor could lose in the event of a default.

Loss given default is the expected amount that would be lost if there is a default. It is calculated using assumed recovery rates.
Probability of default is the risk-neutral default probability. (Risk-neutral default probability > Actual probability of default)
Credit valuation adjustment (CVA) is subtracted from the hypothetical value of the bond, if it were default risk-free, to get the bond's fair value given its credit risk. CVA represents compensation for bearing credit risk.

Structural models predict why a default may occur. They are based on insights from option pricing theory.

Reduced-Form Models predict when a default may occur. They are based on statistical methods. Default is treated as an exogenous variable that occurs randomly.

We can calculate the value of a bond and its credit spread using the following steps:

1. Determine value of bond assuming no default (VND)
2. Calculate credit valuation adjustment (CVA)
3. Fair value of bond = VND - CVA
4. Using fair value, determine YTM.
5. Using YTM, determine credit spread.

Asset backed securities

Three major factors to consider when evaluating asset-backed securities (ABS)

- Underlying collateral
- Origination and servicing of collateral
- Structure of the transaction

Three major credit analysis approaches can be used for ABS:

- Book of loans
- Portfolio
- Loan by loan

Credit default swaps

- Payout ratio = 1 - recovery rate (%)
- Payout amount = payout ratio x notional amount
- Expected Loss = POD x LGD
- Upfront payment = present value of protection leg - present value of premium leg
- Present value of credit spread = upfront premium + present value of fixed coupon
- Upfront premium $\approx$ (credit spread - fixed coupon) x duration
- Credit spread $\approx$ (upfront premium/duration) + fixed coupon
- Price of CDS in currency per 100 par = 100 - upfront premium %
- Upfront premium % = 100 - price of CDS in currency per 100 par
- Profit for the protection buyer $\approx$ change in spread in bps x duration x notional amount
- Percentage change in CDS price = change in spread in bps x duration

Derivatives

Forward contract price

- If underlying has no cash flows: $F = S \times (1 + r)^T$
- If underlying has cash flows: $F = S(1 + r)^T + \text{Future Value of Costs} - \text{Future Value Benefits}$
 $F = \text{Future value of underlying adjusted for carry cash flows} = FV(S_0 + \theta_0 - \gamma_0)$ where θ_0 represents present value of costs at time 0 and γ_0 represents present value of benefits at time 0.
 $V_t = \text{Present value of difference in forward prices}$
- Using continuous compounding: $F_0(T) = S_0 e^{(r_c + \theta - \gamma)T}$

FRA pricing and valuation

FRA fixed rate (i.e. price of an FRA):

1. Set up the time line.

$$2. \text{ Compute the de-annualized fixed rate: } \frac{1 + \left(r_2 \times \frac{n_2}{360} \right)}{1 + \left(r_1 \times \frac{n_1}{360} \right)} - 1$$

$$3. \text{ Annualize by multiplying by } 360 / (n_2 - n_1)$$

FRA value for pay fixed receive floating:

1. Find the new FRA fixed rate at time t
2. Compute payoff at n2: $(FRA_t - FRA_0) \times (n_2 - n_1) / 360 \times NP$
3. Discount back to time t.

Fixed-income forward or futures price:

$$F_0(T) = FV_{0,T}(B_0 + AI_0) - AI_T - FVCI_{0,T}$$

The quoted price which includes the conversion factor is:

$$QF_0(T) = F_0(T) / CF(T)$$

$$V(T) = PV_{t,T}(Ft(T) - F_0(T))$$

Interest rate swaps

Swap fixed rate = $(1 - \text{final discount factor}) / (\text{sum of all discount factors})$

Value at time t = the sum of the present value of the difference in fixed swap rates x notional amount

Currency swaps: two counterparties agree to exchange future interest payments in different currencies. Notional amount of one currency will be given; notional amount of other currency will be based on the spot exchange rate at time 0.

Equity swap pricing and valuation

$$V_t = FB_t(C_0) - \left(\frac{S_t}{S_t^-} \right) NA_E - PV(\text{Par} - NA_E)$$

Binomial model no arbitrage approach (call options)

Hedge ratio, $h = (c^+ - c^-) / (S^+ - S^-)$

$$c = hS + PV(-hS^- + c^-)$$

No-arbitrage approach: a long call option is equal to owning h shares of stock partially financed, where the financed amount is $PV(-hS^- + c^-)$.

Binomial model no arbitrage approach (put options)

Hedge ratio, $h = (p^+ - p^-) / (S^+ - S^-)$

$$p = hS + PV(-hS^- + p^-)$$

$$p = hS + PV(-hS^+ + p^+)$$

Binomial model: expectations approach

Risk-neutral probability of an up move:

$$\pi = (1 + r - d) / (u - d)$$

$$E(c_1) = \pi c^+ + (1 - \pi)c^-$$

$$c = PV_t[E(c_1)]$$

BSM model

without carry benefit:

$$c = SN(d_1) - e^{-rT} XN(d_2)$$

$$p = e^{-rT} XN(-d_2) - SN(-d_1)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

with carry benefit:

$$c = Se^{-\gamma T} N(d_1) - e^{-rT} XN(d_2)$$

$$p = e^{-rT} XN(-d_2) - Se^{-\gamma T} N(-d_1)$$

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r - \gamma + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

	Call option		Put option	
	Underlying	Financing	Underlying	Financing
Binomial model	hS	PV(-hS ⁻ + C ⁻)	hS	PV(-hS ⁻ + p ⁻)
BSM model	N(d ₁)S	-N(d ₂) e ^{-rTX}	-N(-d ₁)S	N(-d ₂) e ^{-rTX}

Alternative Investments

Overview of real estate

Core real estate: Relatively stable, income-producing properties.

Opportunistic real estate investments: More speculative investments, typically funded on a private basis.

Return metric for income-producing property: $\text{NOI} = \text{Effective gross income} - \text{Operating expenses} - \text{Property maintenance allowance}$

$\text{Loan to Value (LTV)} = \text{Mortgage debt outstanding} / \text{Current property value}$

$\text{Debt Service Coverage (DSC)} = \text{Net operating income} / \text{Debt service}$

$\text{Equity dividend rate} = \text{Pre-tax cash flow} / (\text{Property purchase price} - \text{Mortgage loan})$

$\text{Pre-tax cash flow} = \text{Net operating income} - \text{Debt service}$

$\text{After-tax cash flow} = \text{Pre-tax cash flow} - \text{Taxes}$

$\text{Taxes} = t \times (\text{NOI} - \text{Interest expense} - \text{Depreciation expense})$

Real estate operating companies (REOCs): Taxable corporations that own, operate, and manage commercial real estate with few restrictions.

Real Estate Investment Trusts (REITs): Restricted to primarily owning and operating rental properties or purchasing mortgages and are required to distribute nearly all or all of their earnings to investors to avoid paying corporate income tax.

Economic value drivers for real estate: GDP, job, and wage growth; credit conditions contributing to the real estate cycle; and local economic factors.

Phases of real estate cycle: Recovery, Expansion, Oversupply and Recession.

Reasons for investing in RE: Current income, price appreciation, inflation hedge, diversification and tax benefits.

$\text{Gross potential rental income (GPRI)} = \text{Market rent} \times \text{Rentable space}$

Key elements of real estate due diligence: Market Review and Outlook, Current Lease Review, Future Lease Outlook, Financial Review, Documentation Review & Property Inspection and Service Agreements.

Real estate valuation methods include the income, cost and sales comparison approaches.

$\text{Property value} = \text{Expected NOI} / \text{Capitalization rate}$

$\text{Capitalization rate} = r - g$

$\text{Property value} = \sum_{i=1}^n \frac{\text{NOI}_{t+i}}{(1+r)^i} + \frac{\text{Terminal Value}}{(1+r)^n}$

$\text{Terminal value} = \frac{\text{NOI}_n (1+g)}{r-g}$

Main real estate indexes:

Appraisal-based indexes. Disadvantage: volatility gets understated (smoothed); can only compare to other appraised values

Transaction-based indexes. Disadvantage: tend to be "noisy" which leads to random upward or downward movements in the index

$\text{Holding period return} = [\text{NOI} - \text{Capital expenditures} + (\text{Ending market value} - \text{Beginning market value})] / \text{Beginning market value}$

Valuation of REITS

Net asset value approach

- NAV = estimated value of operating real estate + cash and A/R - debt and other liabilities
- NAVPS = NAV / # of shares outstanding
- NAVPS = FFO/share x P/FFO multiple

- NAVPS = FFO/share x P/FFO multiple

$\text{Value of operating assets} = \text{NOI} / \text{cap rate}$

Relative value (price multiple) approach

Major multiples: P/FFO, P/AFFO, EV/EBITDA

$\text{FFO} = \text{Accounting net earnings} + \text{Depreciation expense} + \text{Deferred tax expenses} - \text{Gains (losses) from sales of property and debt restructuring}$

$\text{AFFO} = \text{FFO} - \text{straight line adjustment} - \text{recurring maintenance type capital expenditures and leasing commissions}$

Hedge Fund Strategies

Hedge fund strategies are classified based on a combination of:

- Instrument in which the manager invests
- Trading philosophy
- Types of risks the managers assume

- Equity strategies:** focus on the equity markets; including long/short equity, dedicated short bias and equity market neutral.
- Event-driven strategies:** focus on corporate events, such as mergers and acquisitions, bankruptcy etc.; including merger arbitrage and distressed securities.
- Relative value strategies:** focus on the relative valuation between two or more securities; including fixed-income arbitrage and convertible bond arbitrage.
- Opportunistic strategies:** focus on a multi-asset opportunity set using a top-down approach; including global macro and managed futures.
- Specialist strategies:** focus on niche opportunities that require specialized skill or knowledge; including volatility and reinsurance strategies.
- Multi-manager strategies:** focus on creating a portfolio by combining two or more strategies; including multi-strategy and fund-of-funds.

Hedge fund risk exposures: To analyze the exposure to different risk factors, conditional linear factor models can be used. A basic form of a conditional risk factor model can be expressed as:

$$(\text{Return on HF})_t = \alpha_i + \beta_{i,1}(\text{Factor 1})_t + \beta_{i,2}(\text{Factor 2})_t + \dots + \beta_{i,K}(\text{Factor K})_t + D_i \beta_{i,1}(\text{Factor 1})_t + D_i \beta_{i,2}(\text{Factor 2})_t + \dots + D_i \beta_{i,K}(\text{Factor K})_t + (\text{error})_{i,t}$$

Impact of an allocation to a hedge fund strategy: Hedge funds act as both risk-adjusted return enhancers and diversifiers for the traditional stock/bond portfolio. Adding a 20% allocation of a hedge fund strategy group to a traditional 60%/40% portfolio (for a 48% stocks/32% bonds/20% hedge funds portfolio) typically results in:

- Lower total portfolio standard deviation
- Higher Sharpe and Sortino ratios
- Lower maximum drawdown

Commodities and commodity derivatives

Spot and futures pricing

- Basis = spot price - futures price
- Backwardation (positive basis) and contango (negative basis)

Theories explaining futures returns

Insurance theory: Futures prices are impacted by commodity producers who are long the physical good and are motivated to sell the commodity for future delivery to hedge their production price risk exposure.

The hedging pressure hypothesis: Futures prices are impacted by producers along with consumers who seek to protect themselves

from commodity market price volatility by entering into price hedges to stabilize their projected profits and cash flow.
The theory of storage: Futures prices are impacted by supply and demand dynamics of commodity inventories, including the concept of “convenience yield.” Convenience yield is inversely related to general availability of the commodity.

Futures price = spot price + direct storage costs - convenience yield

Total return on a commodity futures position = price return + roll return + collateral return

Portfolio Management

ETFs rely on a creation/redemption process that is carried out in an OTC primary market between the ETF issuer and authorized participants. This mechanism makes the ETF structure inherently fairer as compared to traditional mutual funds. It also allows an ETF to keep capital gains in check, leading to greater tax efficiency.

ETF premiums and discounts refer to the differences between the exchange price of the ETF and the fund's calculated NAV. The major sources of ETF premiums and discounts are timing differences and stale pricing.

Risks of investing in ETFs include: counterparty risk, fund closures risk, and expectation-related risks.

Primary ETF strategies include: portfolio efficiency, asset class exposure management, and active or factor investing.

ETFs rely on a creation/redemption process that is carried out in an OTC primary market between the ETF issuer and authorized participants. This mechanism makes the ETF structure inherently fairer as compared to traditional mutual funds. It also allows an ETF to keep capital gains in check, leading to greater tax efficiency.

While evaluating how well an ETF tracks its underlying index, a rolling return assessment using periodic performance deviations (tracking differences) are more useful than the standard deviation of daily returns (tracking error).

Arbitrage Pricing Theory:

$$E(R_p) = R_F + \lambda_1 \beta_{p,1} + \dots + \lambda_K \beta_{p,K}$$

Portfolio manager's active return has two components

1. The return from factor tilts: product of the portfolio manager's factor tilts (active factor sensitivities) and the factor returns.
2. The return from security selection: part of active return reflecting the manager's skill in individual asset selection.

Active return: $R_A = R_P - R_B$

Active weight: $\Delta w_i = w_{P,i} - w_{B,i}$

$$R_A = \sum \Delta w_i R_{Ai}$$

Active return = $\sum_{k=1}^K [(\text{Portfolio sensitivity})_k - (\text{Benchmark sensitivity})_k] \times \text{Factor return}_i + \text{Security selection}$

$$\text{Information ratio} = \frac{\bar{R}_P - \bar{R}_B}{s(R_P - R_B)}$$

Active risk squared = active factor risk + active specific risk = risk due to portfolio's different-than-benchmark exposures relative to factors specified in the risk model + risks resulting from the portfolio's active weights on individual assets.

Value at risk is the minimum loss that would be expected a certain percentage of the time over a certain period of time given the assumed market conditions.

Extensions of VaR

- **Conditional VaR (CVaR):** average loss conditional on exceeding the VaR cutoff.
Answers the question: how much can I expect to lose if VaR is exceeded?
Also called: expected tail loss or expected shortfall.
- **Incremental VaR (IVaR):** change in VaR resulting from a change in position size.
VaR with proposed new allocation – VaR with original allocation.
Considers diversification effect.
- **Marginal VaR (Mvar):** Effect of a very small change in a position. Some practitioners interpret ‘small’ as a 1% change.
Considers diversification effect.
- **Relative VaR:** measure of the degree to which the performance of a given investment portfolio might deviate from its benchmark.
Also called ex ante tracking error.

Backtesting

Backtesting: tests a strategy in an actual historical environment, usually over a long period of time. In the rolling-window backtesting methodology, researchers use a rolling-window (or walk-forward) framework, fit/calibrate factors or trade signals based on the rolling window, rebalance the portfolio periodically, and then track the performance over time. Thus, rolling-window backtesting is a proxy for actual investing.

Common mistakes when backtesting

- **Survivorship bias:** drawing conclusions from data that only includes entities that have survived to that point. To avoid this bias, analysts should use point-in-time data (most complete data for any given time period and includes all companies that existed). Survivorship bias is a type of look-ahead bias.
- **Look-ahead bias:** using information that was unknown or unavailable during the historical periods over which the backtest is conducted. Common forms: reporting lags, revisions in data, index additions. Use point-in-time data to avoid this mistake.
- **Data snooping:** making an inference after looking at statistical results rather than testing a prior inference. This can be mitigated by setting higher hurdle rates and cross validation (separating data into training data and testing data).

Methods to supplement backtesting in evaluating new investment strategies

Historical scenario analysis: tests a strategy in a specific historical environment, usually over a short period of time, such as during recessions or periods of high inflation. This can help investors understand the performance of an investment strategy in different structural regimes.

Historical simulation: instead of assuming that a strategy was implemented at some past date and collecting results as the strategy runs over time, we construct results by randomly selecting returns from many different historical periods without regard to time-ordering. It is relatively straightforward to perform but shares pros and cons similar to those of rolling-window backtesting.

Monte Carlo simulation: each key variable is assigned a statistical distribution and observations are drawn at random from that

distribution. The most important decision is the choice of functional form of the statistical distribution of decision variables/return drivers. Multivariate normal distribution is often used in investment research, owing to its simplicity. However, a multivariate normal distribution cannot account for negative skewness and fat tails observed in factor and asset returns.

Sensitivity analysis: a technique for exploring how a target variable and risk profiles are affected by changes in input variables. This can further help investors understand the limitations of conventional Monte Carlo simulation (which typically assumes a multivariate normal distribution as a starting point). A multivariate skewed t-distribution considers skewness and kurtosis but requires estimation of more parameters and thus is more likely to suffer from larger estimation errors.

Inter-Temporal Rate of Substitution

A bond promises \$1 at the end of the year. Inflation rate is 0. The year-end state of the economy is uncertain. Investors are willing to pay more if the economy is expected to perform poorly. If an investor's inter-temporal rate of substitution is higher than the current bond price she will buy.

The one-period real risk-free rate: $l_{t,1} = \frac{1 - P_{t,1}}{P_{t,1}}$

If uncertainty about the state of the economy increases the expected intertemporal rate of substitution will be lower. Investors are compensated with a higher expected return when uncertainty increases.

Default-free interest rates and economic growth

Higher trend real economic growth → higher real default-free interest rates

Higher volatility of GDP growth rate → higher real default-free interest rates

For a real default-free bond we have:

$$P_t^i = \sum_{s=1}^N \frac{CF_{t+s}^i}{(1 + l_{t,s})^s}$$

Only changes in $l_{t,s}$ will affect the price of such a bond. $l_{t,s}$ will be determined by real economic growth and the volatility in economic growth over time as a result of the aggregation of the consumption and saving decisions of individual investors.

Short-term rates and the business cycle

Basic pricing formula for a default-free nominal coupon-paying bond:

$$P_t^i = \sum_{s=1}^N \frac{CF_{t+s}^i}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s})^s}$$

Since T-bills are short-dated, inflation uncertainty can be ignored; they are also zero-coupon bonds, so we do not use the summation term:

$$P_t^i = \frac{CF_{t+1}^i}{(1 + l_{t,1} + \theta_{t,1})^1}$$

Short-term nominal interest rates are positively related to short-term real interest rates and short-term inflation expectations. Central banks set short-term interest rates in response to the economy's position in the business cycle.

- Cut rates when economic activity and/or inflation are judged to be too slow.
- Increase rates when economic activity and/or inflation are judged to be too high.

Taylor rule: $pr_t = l_t + t_t + 0.5(t_t - t_t^*) + 0.5(Y_t - Y_t^*) = l_t + 1.5t_t - 0.5t_t^* + 0.5(y_t - y_t^*)$

Credit premiums and the business cycle

$$P_t^i = \sum_{s=1}^N \frac{E_t[CF_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s}^i)^s}$$

Credit spread = yield on credit-sensitive bond – yield on government bond

- Credit spreads increase in times of economic weakness. Investors demand a return (yield) for possible losses. Expected loss = $PD \times (1 - RR)$.
- Credit spreads narrow as the economy strengthens.
- When spreads narrow, credit-sensitive bonds perform well.
- When spreads narrow relative to government bonds, spreads between higher- and lower-rated bonds also narrow.

Core point: Credit risky bonds perform poorly in bad economic times.

Equities and the equity risk premium

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[CF_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s}^i + K_{t,s}^i)^s}$$

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[CF_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \lambda_{t,s}^i)^s}$$

- Equities do not have good consumption hedging properties (do not pay off in bad times).
- Risk averse investors would demand a higher premium on equity holdings relative to bond holdings.
- ERP and CRP are positively correlated and are influenced by the business cycle in similar ways.

Commercial real estate

$$P_t^i = \sum_{s=1}^{\infty} \frac{E_t[CF_{t+s}^i]}{(1 + l_{t,s} + \theta_{t,s} + \pi_{t,s} + \gamma_{t,s}^i + K_{t,s}^i + \phi_{t,s}^i)^s}$$

- Commercial real estate has bond-like and equity-like characteristics. Rental income is like coupon payments. Value of property at the end of a lease term is uncertain.
- Commercial real estate is generally illiquid.
- Commercial real estate prices are pro-cyclical.
- Investors will demand a higher risk premium.

Decomposition of value added

$RA = (\Delta w_{stocks} R_{B,stocks} + \Delta w_{bonds} R_{B,bonds}) + (w_{P,stocks} R_{A,stocks} + w_{P,bonds} R_{A,bonds})$ = Value added due to asset allocation + Value added due to security selection

Information Ratio (IR) is affected by the addition of cash or leverage in a portfolio. Adding cash to a portfolio of risky assets will cause the IR to decrease. The IR of an unconstrained portfolio is unaffected by the aggressiveness of active weights. If the active security weights are all multiplied by some constant, c , the information ratio of an actively managed portfolio will remain unchanged.

$$IR_C = c \cdot RA / c \cdot STD(RA) = RA / STD(RA)$$

Constructing optimal portfolios

Optimal portfolio (P) should have the highest possible Sharpe ratio $SR_P^2 = SR_B^2 + IR^2$

Optimal amount of active risk: $STD(R_A) = STD(R_B) \times (IR / SR_B)$

Optimal portfolio (P): $STD(R_P)^2 = STD(R_B)^2 + STD(R_A)^2$

The Full Fundamental Law

$$E(R_A) = (TC) (IC) \sqrt{BR} \sigma_A$$

$$IR = (TC) (IC) \sqrt{BR}$$

$$TC = COR(\mu_i/\sigma_i, \Delta w_i \sigma_i)$$

$$\sigma_A = TC \frac{IR}{SR_B} \sigma_B$$

$$SR_P^2 = SR_B^2 + (TC)^2 (IR^*)^2$$

For unconstrained portfolios, IR does not change with active risk.

For constrained portfolios, IR generally decreases as the active risk increases.

Ethical and Professional Standards

I(A) Knowledge of the law: comply with the strictest law; disassociate from violations.

I(B) Independence and objectivity: do not offer, solicit or accept gifts; but small token gifts are allowed.

I(C) Misrepresentation: do not guarantee performance; avoid plagiarism.

I(D) Misconduct: do not behave in a manner that affects your professional reputation or integrity.

I(E) Competence: act with and maintain the competence necessary to fulfill professional responsibilities.

II(A) Material nonpublic information: do not act or help others to act on this information; mosaic theory is not a violation.

II(B) Market manipulation: do not manipulate prices/trading volumes to mislead others; do not spread false rumors.

III(A) Loyalty, prudence, and care: place client's interest before employer's or your interests.

III(B) Fair dealing: treat all clients fairly; disseminate investment recommendations and changes simultaneously.

III(C) Suitability: in advisory relationships, understand your client's risk profile, and develop and update an IPS periodically; in fund/index management, ensure investments are consistent with the stated mandate.

III(D) Performance presentation: do not misstate performance; make detailed information available on request.

III(E) Preservation of confidentiality: maintain confidentiality of clients (unless disclosure is required by law, information concerns illegal activities, or client permits the disclosure).

IV(A) Loyalty: do not harm your employer; obtain written consent before starting an independent practice; do not take confidential information when leaving.

IV(B) Additional compensation arrangements: do not accept compensation arrangements that will create a conflict of interest with your employer, but you may accept if written consent is obtained from all parties involved.

IV(C) Responsibilities of supervisors: prevent employees under your supervision from violating laws.

V(A) Diligence and reasonable basis: have a reasonable and adequate basis for any analysis, recommendation or action.

V(B) Communication with clients and prospective clients: distinguish between fact and opinion; make appropriate disclosures.

V(C) Record retention: maintain records to support your analysis.

VI(A) Avoid or disclose conflicts: avoid or make full and fair disclosure of all matters that could impair independence and objectivity; disclosures should be made in plain language.

VI(B) Priority of transactions: client transactions come before employer transactions which come before personal transactions.

VI(C) Referral fees: disclose referral arrangements to (current and prospective) clients and employers.

VII(A) Conduct as participants in CFA Institute programs: don't cheat on the exams; keep exam information confidential.

VII(B) Reference to CFA Institute, the CFA designation, and the CFA program: don't brag; references to partial designation are not allowed.

Other IFT products to help you ace the Level II exam:

- Detailed videos
- High-Yield videos
- High-Yield course
- Study notes
- Question bank
- Mock exams
- Study strategy and advice

Can be found at www.ift.world

Click for more details and free stuff

